FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2024

TEST DATE: 07-01-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. let the scooterist velocity be v. Then

$1000+(10\times100)=v\times100$

$\Rightarrow 100v=2000\Rightarrow v=\frac{2000}{100}20\,m/s$

2. B

3. B

Sol. In the given equation $[\rho]=[b][x]$;

$\therefore [b]=[\rho]/[x]$. But ρ is mass per unit length and x is distance, therefore

$[b]=ML^{-1}/L=ML^{-2}T^{0}$.

4. D

Sol. Point A is at rest w.r.t. motion, hence, v at A = 0. At point B there are two horizontal velocities, hence $v_B = 2v$

5. D

Sol. $mg = 2TL \Rightarrow \pi r^2 Ldg = 2TL \Rightarrow \Rightarrow \pi r^2 dg = 2T$.

This relation is independent of L.

6. D

7. D

8. C

Sol. $\theta = \dfrac{K_1\phi_1 l_2 + K_2\phi_2 l_1}{K_1 l_2 + K_2 l_1} = \dfrac{K\times 0\times 2 + 3K\times 100\times 1}{K\times 2 + 3K\times 1} = 60°C$

9. C

Sol. $\dfrac{1}{C_\infty} = \dfrac{1}{C_\infty + C} + \dfrac{2}{C} = \dfrac{3C + 2C}{C_\alpha\left(C_\alpha + C\right)}$

$2C_\infty^2 + 2CC_\infty - C^2 = 0 \rightarrow C_\infty = C\left(\dfrac{-1+\sqrt{3}}{2}\right)$

10. B

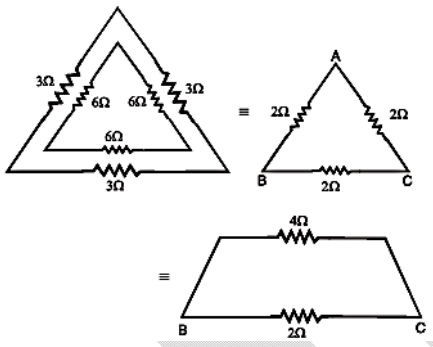
Sol. R increasing with increasing temperature.

V = IR

Slope of graph = I/V = 1/R; Slope of T_1 is more i.e. $1/R_1$ is more, hence R_1 is less. This concludes that T_1 will be less than T_2 as R_1 is less than R_2.

11. C

Sol. $R_{eq} = \dfrac{R_1R_2}{R_1+R_2}\left(parallel\right) = \dfrac{4\times 2}{4+2} = \dfrac{8}{6} = \dfrac{4}{3}\Omega$



12. D

Sol. $i_R = \dfrac{V_0}{R} = \dfrac{100}{20} = 5,\ i_L = \dfrac{V_0}{X_L} = \dfrac{100}{10} = 10\ and\ i_C = \dfrac{V_0}{X_C} = \dfrac{100}{20} = 5$

Current, $i = \sqrt{i_R^2 + \left(i_C - i_L\right)^2} = \sqrt{5^2 + 5^2} = 5\sqrt{2}\ amp.$

13. A

Sol. $f_{\max} = \mu mg,\ a_{\max} = \mu g$

If A is the amplitude $a_{\max} = A\omega^2 = 4\pi^2 A\upsilon^2 = \mu g$

Therefore, $A = \dfrac{\mu g}{4\pi^2\upsilon^2}$

14. C

Sol. Total time taken to travel distance d is:

$\dfrac{d}{2n_1} + \dfrac{d}{2n_2} = d\left(\dfrac{n_1+n_2}{2n_1n_2}\right) = \dfrac{d}{n_{eff}};\ n_2 = 3n_1 \Rightarrow n_{eff} = \dfrac{3}{2}n_2$

15. B

Sol. $(A_1 + A_2) + (A_3 + A_4) = A_{12} + A_{34}$

16. A

Sol. $mvr = \frac{nh}{2\pi}, \lambda = \frac{h}{mv};$

$2\pi r = \frac{1 \times h}{mv} .....(i)$

$\lambda = \frac{h}{mv} .....(ii)$

Divide (ii) by (i), $\frac{2\pi r}{\lambda} = \frac{h \times mv}{mv \times h} = \frac{1}{1} = 1:1$

17. A

Sol. $V_B - V_A = -\left[\int_1^2 2dx + \int_2^1 3dy\right]$

$= -\left[2(2-1) + 3(1-2)\right]$

$= -\left[2-3\right] = 1V$

$V_A - V_B = -1\text{V}$

18. D

Sol. In the graph given, slope of curve 2 is greater than the slope of curve 1.

$\left(\frac{\gamma P}{V}\right)_2 > \left(\frac{\gamma P}{V}\right)_1 \Rightarrow \gamma_2 > \gamma_1$

$\gamma_{He} > \gamma_{O_2}$

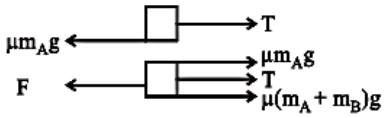
19. B

20. B

SECTION – B

21. 10

Sol. Here, m_A = 0.5 kg; m_B = 1kg



Force on block A

$T = \mu m_A g$ ……... (1)

Force acting on block B

$F = T + \mu m_A g + \mu(m_A + m_B)g$ ……. (2)

From (1) and (2)

$F = \mu m_A g + \mu m_A g + \mu m_A g + \mu m_B g$

$F = 3\mu m_A g + \mu m_B g = \mu g(3m_A + m_B)$

$= 0.4 \times 10 \times (3 \times 0.5 + 1) = 10N$

22. 1

Sol. Given, $v_0 = \frac{v_e}{2}$

$$\sqrt{\frac{GM}{R+h}} = \frac{1}{2}\sqrt{\frac{2GM}{R}}$$

$$h = R$$

When the satellite stops, K = 0

$$-\frac{GMm}{2R} - \left(-\frac{GMm}{R}\right) = \frac{1}{2}mv^2$$

$$v = \sqrt{gR}$$

23. 325

Sol. $n_1 C_{v_1} T_1 + n_2 C_{v_2} T_2 = (n_1 + n_2) C_v T$

$$C_{v_1} T_1 + C_{v_2} T_2 = 2C_v T \left(\text{as } n_1 = n_2\right)$$

$$C_v = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2} = 2R$$

$$T = \frac{\frac{3}{2}R \times 200 + \frac{5}{2}R \times 400}{2 \times 2R} = 325$$

24. 2

Sol. Magnetic induction at O due to coil Y is given by,

$$B_Y = \frac{\mu_0}{4\pi} \times \frac{2\pi I (2r)^2}{\left[(2r)^2 + d^2\right]^{3/2}} \ldots\ldots\ldots\ldots (1)$$

Similarly, the magnetic induction at O due to coil X is given by

$$B_X = \frac{\mu_0}{4\pi} \times \frac{2\pi I r^2}{\left[r^2 + (d/2)^2\right]^{3/2}} \ldots\ldots\ldots\ldots (2)$$

From eq. (1) & (2)

$$\frac{B_Y}{B_X} = \frac{1}{2}$$

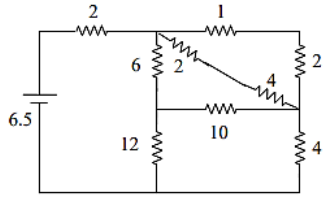
25. 9

Sol. Field inside the shell is zero, so potential is uniform.

$$V = \frac{kq}{2R} = 9 \text{ kilo volts}$$

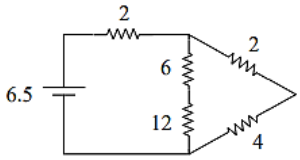
26. 1

Sol. Due to balanced wheatstone bridge ABCEA BE=8Ω can be taken out.

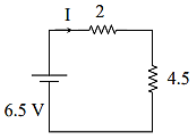


$$R_{AC}=\frac{(1+2)(2+4)}{(1+2)+(2+4)}=\frac{18}{9}=2$$

Now, ACFGHA is a balanced Wheatstone bridge, hence HC=10Ω can be taken out



$$R_{AG}=\frac{6\times18}{6+18}=\frac{108}{24}=4.5\Omega$$



Hence, current $i=\frac{V}{R_{total}}=\frac{6.5}{6.5}=1A$

27. 9

28. 4

29. 2

Sol. Given that:

$x=40\cos(50\pi t-0.02\pi y)$

Particle velocity

$v_p=\frac{dx}{dt}=(40\times50\pi)(-\sin(50\pi t-0.02\pi y))$

Putting y = 25 cm and t = 1/200 s

$v_p = 10\pi\sqrt{2}$

30. 0

Sol. All the four corners will be at the same potential.

$W = q\Delta V = 0$

Chemistry

PART – B

SECTION – A

31. A

Sol. $C_6H_5NH_2 + H_2O \rightleftharpoons C_6H_5NH_3^+ + OH^-$

1 0 0

$1-\alpha$ α α

(∵ Dissociation occurs in presence of NaOH and thus dissociation of $C_6H_5NH_2$ will suppress)

$$K_b = \frac{[C_6H_5NH_3^+][OH^-]}{[C_6H_5NH_2]} = \frac{10^{-8} \text{ x } [OH^-]}{[0.24]}$$

∵ K_b for $C_6H_5NH_2 = \frac{K_w}{K_a \text{ for } C_6H_5NH_3^+} = \frac{10^{-14}}{2.4 \text{ x } 10^{-5}}$

$$\therefore \quad [OH^-] = \frac{10^{-14} \text{ x } 0.24}{2.4 \text{ x } 10^{-5} \text{ x } 10^{-8}} = 0.01.$$

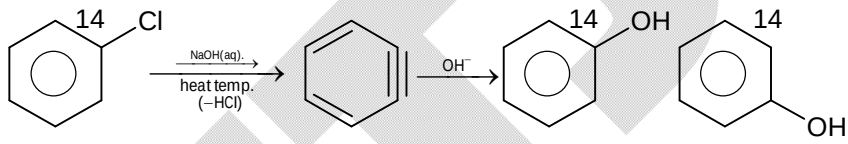
∴ NaOH = 10^{-2} M

32. A

Sol. $K = \text{antilog}\left[\frac{nE^o}{0.059}\right] = \text{antilog}\left[\frac{2x(-0.295)}{0.059}\right] = 1x10^{10}$

33. C

Sol.



34. B

Sol. $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightarrow NH_3(g)$

$$\Delta H_{298K} = \sum (H_f)_p - \sum (H_F)_R = -(11.02 - 0) = -11.02\, kcal\, mol^{-1}$$

$$\frac{\Delta H_2 - (-11.02)}{T_2 - T_1} = \Delta C_p$$

$$\frac{\Delta H_2 - (-11.02)}{773 - 298} = \left(8.38 - \frac{1}{2} \times 6.96 - \frac{3}{2} \times 6.89\right) \times 10^{-3}$$

$$\Delta H_2 = -13.6\, kcal\, mol^{-1}$$

35. C

Sol. This compound has only one ionizable Cl^- out of given three

$[Co(NH_3)_4Cl_2]Cl \rightleftharpoons [Co(NH_3)_4Cl_2]^+ + Cl^-$

36. B

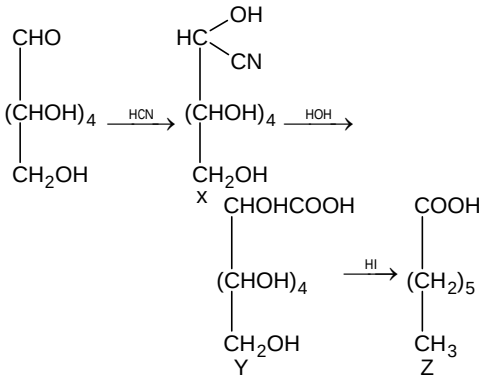
Sol. $\left[Cr(H_2O)_6\right]^{2+}$ had Cr^{2+}

$Cr^{2+} = 3d^4 4s^0$ (four unpaired electrons)

$Fe^{2+} = 3d^6 4s^0$ (in the presence of weak ligand, it also has four unpaired electrons).

37. C

Sol.



38. B

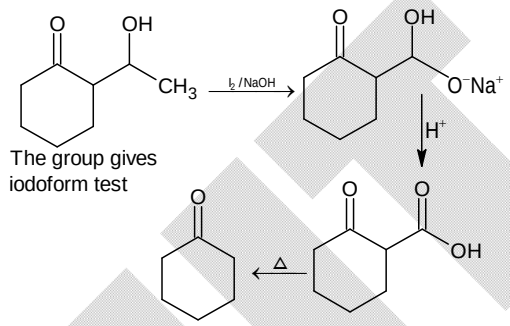
Sol. Diazonium ion acts as an electrophile in coupling reaction. The greater the electron- withdrawing power, the higher is the electrophilicity.

39. B

Sol. $BrCH_2CH_2CHO \xrightarrow[HCl]{Zn(Hg)} BrCH_2CH_2CH_3$

40. A

Sol.



The group gives iodoform test

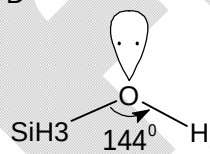
If keto group is at β-position w. r. t. COOH heating causes decarboxylation

41. A

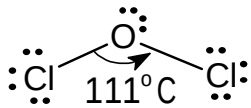
Sol. On cis-alkene there is syn addition of two-OH groups forming meso compound

42. D

Sol.



Hybridisation of O-atom sp^2 due to more extent of $(2p_\pi - 3d_\pi)$ back bonding



Hybridisation of O-atom sp^3, because of less effective $(2p_\pi - 3d_\pi)$ back bonding due to d-orbital resonance

43. D

Sol. $Cu^{2+}(aq) + 4NH_3 \rightleftharpoons^{OH^-} \left[Cu(NH_3)_4\right]^{2+}$

$Cu^{2+}(aq) + 4NH_3 \overset{H^+}{\rightleftharpoons} Cu^{2+} + 4NH_4^+ (No\ complex\ formation)$

As NH_3 combines with H^+ of acid the changes to NH_4^+ which have no doner site.

44. B

Sol. $Zn^{++} + 2H_2O \rightleftharpoons Zn(OH)_2 + 2H^+$

$$K_h = \frac{\left[Zn(OH)_2\right]\left[H^+\right]^2}{\left[Zn^{++}\right]} \ldots\ldots\ldots\ldots\ldots(1)$$

$$K_b = \frac{\left[Zn^{++}\right]\left[OH^-\right]^2}{\left[Zn(OH)_2\right]},\ K_w = \left[H^+\right][OH]$$

$$\therefore \frac{K_w^2}{K_b} = K_h$$

45. D

Sol. order of electron gain enthalpy Cl > F > O

Second electron gain enthalpy for an element is always positive.

46. A

47. D

Sol. $\ddot{P}H_3 + H^+ \rightarrow PH_4^+$

According to Drago's rule lone pair on phosphorus resides in almost pure s-orbital, hence due to non-directional nature, its overlapping tendency is greatly reduced in comparison to a lone pair present in hybrid orbital, which is directional as present in $\ddot{N}H_3$.

48. B

Sol. $NaCl_{(s)} + K_2Cr_2O_{7(s)} + H_2SO_4\ (few\ drops) \xrightarrow[reaction]{Non-redox} CrO_2Cl_2\ (orange\ red\ vapours)\uparrow + HO$

$CrO_2Cl_2 + OH^-_{(eq.)} \rightarrow CrO^{2-}_{4(aq.)}\ (yellow\ sol.) + Cl^- + 2H_2O$

49. B

Sol. Stability order is I > II > III > IV, because I is neutral, when in II all atoms with complete octet system while is III is more stable than IV because in case of IV oxygen is positive with incomplete octet system.

50. B

Sol. For $M^+ + X^- \rightarrow M + X$, $E^\circ_{cell} = 0.44 - 0.33 = 0.11$ is positive spontaneous.

SECTION – B

51. 33

Sol. ($\because$ Moles $\propto$ volume at constant P and T)

$$N_2O_4 \rightleftharpoons 2NO_2(g)$$

Initial Moel a 0

Mole at equilibrium $(a-x)$ 2x

Given original volume = (75/100) or existing volume at equilibrium

Or Initial volume = (75/100) x mole at equilibrium

$\therefore \quad a = \frac{75}{100}(a + x)$

or $x = \frac{25}{75}a = 0.33a$

$\therefore$ Degree of dissociation, $\alpha = \frac{x}{a}$

$= \frac{0.33a}{a} = 0.33$ or 33%

52. 2

Sol. The ratio constant for the first-order reaction is given by

$k = \frac{2.303}{t}\log\frac{a}{a-x}$

Now, when three-fourth of the reaction is complete,

$x = \frac{3a}{4}$

$\therefore \quad t_{3/4} = \frac{2.303}{k}\log\frac{a}{a-3a/4} = \frac{2.303}{k}\log 4$ ...(1)

and when the reaction is half completed, x = a/2

$\therefore \quad t_{1/2} = \frac{2.303}{k}\log\frac{a}{a-a/2} = \frac{2.303}{k}\log 2$ ...(2)

Dividing (1) by (2), we get

$\frac{t_{3/4}}{t_{1/2}} = \frac{\log 4}{\log 2} = \frac{0.6020}{0.3010} = 2$

Thus, the time taken for three-fourth of the reaction to be completed is double than that taken for half of the reaction

53. 28

Sol. The energy needed to see object = 10^{-17} J

Photon energy used to see object = $\frac{hc}{\lambda}$

$= \frac{6.626 \text{ x } 10^{-34} \text{ x } 3.0 \text{ x } 10^{8}}{550 \text{ x } 10^{-9}}$

$= 3.61 \text{ x } 10^{-19}$ J/Photon

$\therefore$ Number of photons = $\frac{10^{-17}}{3.61 \text{ x } 10^{-19}} = 27.7$

$\therefore$ Minimum number of photons required to see object = 28

54. 1

Sol. $t_{1/2} \propto (a_0)^{1/(n-1)}$; where n = order of reaction

$t_{1/2} \propto (a_0)^{\frac{1}{\left(-1+\frac{3}{2}\right)}}$ .....(1)

$t_{1/2} \propto (a_0)^{m}$ .....(2)

$m = \frac{3}{2} - 1 \Rightarrow m = 0.5$

$2m = 1$

55. 4

Sol. $\log \frac{k_2}{k_1} = \frac{E}{2.303}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$

$\log 2 = \frac{E}{2.303}\left[\frac{10}{290 \times 300}\right]$

$\therefore E = 12062\, cal \approx 12\, kcal$

56. 9

Sol. Total energy released during combustion of 3.5 g gas = (mS) x ΔT

= 2.5 x 0.45 = 1.125 kJ

$\therefore$ Heat released by 1 mole of gas combustion.

= $\frac{28}{3.5}$ x 1.125 (molar mass of gas = 28)

= $9\, kJ\, mol^{-1}$

57. 3

Sol.

HA	$\rightleftharpoons$	H^+	+	A^-
0.1		0		0
$0.1\left(1-\frac{1}{100}\right)$		$\frac{1\times 0.1}{100}$		$\frac{1\times 0.1}{100}$

$\left[H^+\right] = 10^{-3} \therefore pH = 3$

58. 66

Sol. The redox changes are

$Mn^{7+} + 5e \rightarrow Mn^{2+}$

$\left(C^{3+}\right)_2 \rightarrow 2C^{4+} + 2e$

$\because$ meq. of oxalate ion = meq. of $KMnO_4$

(w/E) x 1000 = 90 x (1/20)

$\frac{w}{88/2}$ x $1000 = \frac{9}{2}\left(\because E_{C_2O_4^{2-}} = \frac{88}{2}\right)$

$w_{C_2O_4^{2-}}$ = 0.198g

$\therefore$ % of oxalate in sample = (0.198 x 100)/0.3 = 66%

59. 136

Sol. $\Delta T = 5.51 - 1.59 = 3.92$, w = 0.186 g

W = 7.5g, $k_f = 4.9$ k molality^{-1}

$\because \quad \Delta T = \frac{1000 \text{ x } K_f \text{ x } w}{m \text{ x } W}$

$\therefore 3.92 = \frac{1000 \text{ x } 4.9 \text{ x } 0.186}{m \text{ x } 7.5}$

$\therefore$ m = 136 g mol^{-1}

60. 10

Sol. $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

+7 +6 +8 +2

$2MnO_4^- \equiv 5C_2O_4^{2-}$

$\frac{M_1V_1}{2} = \frac{M_2V_2}{5}$

$\frac{0.2 \times V_1}{2} = \frac{0.1 \times 50}{5}$

$V_1 = 10mL$

Mathematics

PART – C

SECTION – A

61. D
Sol. $a^2 + 7a + 1 < 0$ and $a^2 + 8a + 4 < 0$ which satisfied by 6 negative integers

62. A
Sol. Relation R is reftexive and symmetry as xRy is true then yRx is also true

63. B
Sol. Any number having exactly 4 factors is of the form $m = p^3$ (p prime) or m = p.q (where p & q are distinct primes)
So we have ${}^5C_2 + 2 = 12$ such factors

64. B
Sol. $\lim\limits_{x\to\infty}[x] = \lim\limits_{x\to\infty}\dfrac{[x]x}{x}$

$= \lim\limits_{x\to\infty}\dfrac{[x]x}{x}.\lim\limits_{x\to\infty}x = \lim\limits_{x\to\infty}x\left(\text{as }\lim\limits_{x\to\infty}\dfrac{[x]}{x}=1\right)$

$= \lim\limits_{x\to\infty}\dfrac{x^n+x^{n-1}+1}{e^{[x]}}$

$= \lim\limits_{x\to\infty}\dfrac{x^n+x^{n-1}+1}{e^{x}}$

$= \lim\limits_{x\to\infty}\dfrac{1+\dfrac{n}{x}+\dfrac{1}{x^n}}{1+x+\dfrac{x^2}{2!}+..+\dfrac{x^n}{n!}+\dfrac{x^{n+1}}{(n+1)!}+...}$

$= \lim\limits_{x\to\infty}\dfrac{1+\dfrac{n}{x}+\dfrac{1}{x^n}}{\dfrac{1}{x^n}+\dfrac{1}{x^{n-1}}+\dfrac{1}{2x^{n-2}}+....+\dfrac{1}{n!}+\dfrac{x}{(n+1)!}+\dfrac{x^2}{(n+2)!}+...}$

$= \dfrac{1+0+0}{0+0+..+0+\dfrac{1}{n!}} = \dfrac{1}{\infty} = 0..$

65. A
Sol. The given equation is $dx - x(ydx + xdy) = x^5y^4(ydx + xdy)$

$\Rightarrow \dfrac{dx}{x} = (1+x^4y^4)d(xy) \Rightarrow \ln x = xy + \dfrac{1}{5}x^5y^5 + \ln c$

$\Rightarrow x = ce^{xy+\frac{1}{5}x^5y^5}$

66. B
Sol. The distance of the point (1,3) from the line 3x+4y=5 is 2. Now the value of $\sec^2\theta + 2\text{cosec}^2\theta$ is greater than 2 hence two such lines will be possible.

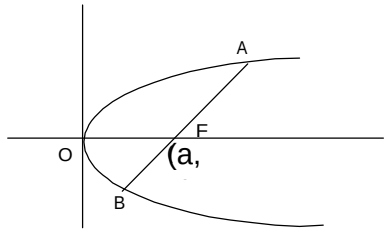
67. B
Sol. Incircle and circumcircle have same centre but radius is half of that of circumcircle
Also $c = -(2g + 2f + 2)$ as (1, 1) lies on circumcircle

68. A

Sol. FA = 4, FB = 5

We know that $\frac{1}{a}=\frac{1}{AF}+\frac{1}{FB}$

$\Rightarrow$ a = $\frac{20}{9}\Rightarrow 4a=\frac{80}{9}$



69. C

Sol. For the given ellipse a^2 = 16. Therefore e = $\sqrt{1-\frac{b^2}{a^2}}$

$\Rightarrow$ e = $\sqrt{1-\frac{b^2}{a^2}}=\sqrt{\frac{16-b^2}{4}}$

So, the foci ellipse are $(\pm ae,0)$ i.e. $\left(\pm\sqrt{16-b^2},0\right)$

For the hyperbola e is given by

e = $\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1+\frac{81}{144}}=\frac{15}{12}=\frac{5}{4}$

So, $16-b^2=\left(\frac{12}{5}x\frac{5}{4}\right)^2=9\Rightarrow b^2=7$

70. B

Sol. $\cos^2\theta=\frac{a^2-1}{3},\tan\frac{\theta}{2}=\tan^{2/3}\alpha$

$\tan^3\frac{\theta}{2}=\tan\alpha\Rightarrow\frac{\sin^3\frac{\theta}{2}}{\cos^3\frac{\theta}{2}}=\frac{\sin\alpha}{\cos\alpha}$

$\frac{\sin^3\frac{\theta}{2}}{\sin\alpha}=\frac{\cos^3\frac{\theta}{2}}{\cos\alpha}=k$

$\sin^3\frac{\theta}{2}$ = k sin α ...(I)

$\cos^3\frac{\theta}{2}$ = k cos α ...(II)

$k^{2/3}\sin^{2/3}\alpha + k^{2/3}\cos\alpha = 1$

$\sin^{2/3}\alpha + \cos^{2/3}\alpha = \frac{1}{k^{2/3}}$

Squaring and adding (I) and (II)

$k^2(\sin^2\alpha+\cos^2\alpha)=\sin^6\frac{\theta}{2}+\cos^6\frac{\theta}{2}=\left(\sin^2\frac{\theta}{2}+\cos^2\frac{\theta}{2}\right)^3-3\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}\left(\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}\right)$

$k^2=\frac{a^2}{4}\Rightarrow k=\frac{a}{2}$

$\sin^{2/3}\alpha + \cos^{2/3}\alpha = \left(\frac{2}{a}\right)^{2/3}$

71. C

Sol. Let assume that 'A' wins after n deuces, $n \in [(0,\infty)]$

Probability of a deuce = $\frac{2}{3}.\frac{2}{3}+\frac{1}{3}.\frac{1}{3}=\frac{5}{9}$

(A wins his serve then B wins his serve or A loses his serve then B also loses his serve) Now probability of 'A' winning the game

$\sum_{n=0}^{\infty}(5/9)^n.\left(\frac{2}{3}\right)\frac{1}{3}=\frac{1}{1-(5/9)}.\frac{2}{9}=\frac{1}{2}$

72. A

Sol. $n(A) = 31x31$

$n(B) = 41 \times 21$

$n(C) = 11 \times 11x\ 11$

$n(A \cap B) = 31 \times 21$

$n(B \cap C) = 11 \times 11$

$n(C \cap A) = 11 \times 11$

$n(A \cap B \cap C) = 11 \times 11$

So $n(A \cup B \cup C)$

= 2381

73. A

Sol. 3 median = 2 Mean +Mode

74. D

Sol. There will be $(n_1 + 1)$ gaps created by n_1 men. Now women have to be seated only in these gaps.

Thus number of such sitting arrangements = ${}^{n_1+1}C_{n_2}\ (n_1!)(n_2!)$

75. C

Sol. Since $x^2y^3z^4$ is occurring in the expansion of $(x +y +z)n$, so n should be 9 only.

Now A = $\frac{9!}{2!x3!x4!}=1260$

Coefficient of $x^4\ y^4\ z$ = $\frac{9!}{4!x4!}=630=\frac{A}{2}$

76. B

Sol. $\frac{xy}{x+y}=n \Rightarrow xy-nx-ny=0$

$\Rightarrow (x - n)\ (y - n) = n^2$

No. of positive solutions = No. of division of n^2

= 7 x 9 x 13 = 819

77. D

Sol. Let = $I=\int\frac{(1+x)}{x(1+xe^x)^2}dx=\int\frac{(1+x)e^x}{(xe^x)(1+xe^x)^2}dx(1+xe^x=p, e^x(1+x)dx=dp)$

$I=\int\frac{dp}{(p-1)p^2}$

$\frac{dp}{(p-1)p^2}=\frac{A}{(p-1)}+\frac{B}{p}+\frac{C}{p^2}$

$1 = Ap^2 + B(p)\ (p-1)+C(p-1)$

For = p = 1, p = 0, and p = -1, A = C, C = -1 and B = -1

$\Rightarrow I=\int\frac{1}{(p-1)}dp-\int\frac{dp}{p}-\int\frac{dp}{p^2}=\ln\frac{(p-1)}{p}+\frac{1}{p}+c=\ln\left(\frac{xe^x}{1+xe^x}\right)+\left(\frac{1}{1+xe^x}\right)+c$

78. A

Sol. $\lim\limits_{x\to\infty}\frac{1}{n}\left(\sum\limits_{r=1}^{n}\tan\left(\frac{r\pi}{3n}\right)\right)$

$\frac{r}{n}=x,\frac{1}{n}=dx$

$\int\limits_0^{\pi/3}\tan\left(\frac{x\pi}{3}\right)dx=\frac{3}{\pi}|\ln\sec t|_0^{\pi-3}=\frac{3}{\pi}\ln 2$ $\qquad (\because t=\pi x/3)$

79. B

Sol. $\because\ \vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

$\therefore\ \lambda\left(\vec{a}\times\vec{b}\right)=2\left(\vec{b}\times\vec{c}\right)$

$\Rightarrow\ \lambda=2$

80. B

Sol. $\vec{a}.\vec{b}=(x\hat{i}+(x-1)\hat{j}+\hat{k}).((x+1)\hat{i}+\hat{j}+a\hat{k})$

$= x\ (x + 1) + x - 1 + a$

$= X^2 + 2x + a - 1$

We must have

$\vec{a}.\vec{b}>0\forall x\in R$

$\Rightarrow\ x^2 + 2x + a - 1 > 0\forall x\in R$

$\Rightarrow\ 4-4\ (a-1) < 0$

$\Rightarrow\ a > 2$

SECTION – B

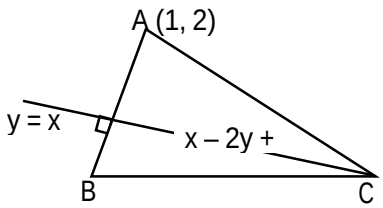
81. 2

Sol. Image of A(1, 2) in y = x will be So B is (2, 1)

Image of A(1, 2) in $x - 2y + 1 = 0$ will lie on BC.

So another point on BC is $\left(\frac{9}{5},\frac{2}{5}\right)$

Equation of BC is $3x-y-5=0$



82. 2

Sol. a, b, c are in H.P $\Rightarrow b=\frac{2ac}{a+c}\Rightarrow\frac{b}{a}=\frac{2c}{a+c}\Rightarrow\frac{b+a}{b-a}=\frac{3c+a}{c-a}$ ...(i)

Again a, b, c are in H.P.

$$b=\frac{2ac}{a+c}\Rightarrow\frac{b}{a}=\frac{2a}{a+c}\Rightarrow\frac{b+c}{b-c}=\frac{3a+c}{a-c} \quad \ldots(ii)$$

From (A) and (B) $\frac{b+a}{b-a}+\frac{b+c}{b-c}=\frac{3c+a}{c-a}+\frac{3c+a}{a-c}=2$

83. 0

Sol. The combined equation of the tangents drawn from (0,0) to $x^2+y^2-2rx-2hy+h^2=0$ is

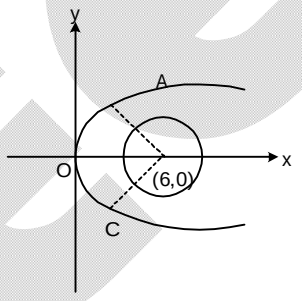
$(x^2+y^2-2rx-2hy+h^2)h^2=(-rx-hy+h^2)^2$

This equation represents a pair of perpendicular straight lines

If coefficient of x^2 + coefficient of $y^2=0$ i.e. $2h^2-r^2-h^2=0\Rightarrow r^2=h2$ or $r=\pm h$.

84. 5

Sol. Shortest distance will take place through along the common normal



Equation of normal for $y^2=4x$ at $(at^2, 2t)$ is

$y=-tx+2t+3t$

If it passes through (6, 0),

then $t^3-4t=0$

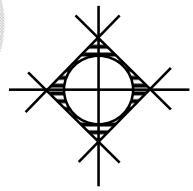
$\Rightarrow$ $t=0, t=\pm 2$

$\therefore$ Min distance = $\sqrt{20}-\sqrt{25}$

$=2\sqrt{5}-\sqrt{5}=\sqrt{5}$

85. 6

Sol. Shaded region is the required one.



$\therefore$ Required Area = $4x\frac{1}{2}x2x2-\pi.2=8-2\pi$ sq. units

86. 21

Sol. Hint: trace of $\left(A_n^{-1}\right)=\frac{1}{a_{11}}+\frac{1}{a_{22}}+\frac{1}{a_{33}}+\ldots$

$$A_3=\begin{bmatrix}a_1 & b_1 & c_1\\ 0 & b_2 & c_2\\ 0 & 0 & c_3\end{bmatrix}$$

$$\text{then } A^{-1}=\begin{bmatrix}\frac{1}{a_1} & k_1 & k_2\\ 0 & \frac{1}{b^2} & k_3\\ 0 & 0 & \frac{1}{c_3}\end{bmatrix}$$

$\therefore$ $\frac{1}{10}\left(\sum_{n=1}^{5}\text{trace of}\left(A_n^{-1}\right)\right)$

$=\frac{1}{10}\left(\text{tr}(A_1^{-1})+\text{tr}(A_2^{-1})+\text{tr}(A_3^{-1})+\text{tr}(A_4^{-1})+\text{tr}(A_5^{-1})\right)$

$= \frac{1}{10}(5x2+4x8+3x18+2x32+50)$

$= 21$

87. 63

Sol. When everyone gets a different number = ${}^6C_4.\underline{|4}$

When only two persons siting opposite to each other get same number = ${}^6C_3.{}^3C_1.2.\underline{|2}$

When two pairs of persons sitting opposite to each other get same number ${}^6C_2.\underline{|2}$

So = N = 630

88. 11

Sol. 15 $|\overrightarrow{AC}| = 3|\overrightarrow{AB}| = 5|\overrightarrow{AD}|$

Given

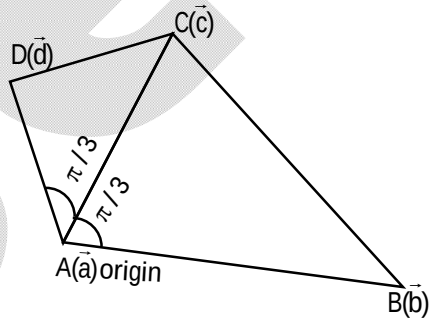
Let $|\overrightarrow{AC}| = \lambda \neq 0$

$\therefore \quad |\overrightarrow{AB}| = 5\lambda$

$|\overrightarrow{AD}| = 3\lambda$

Now $\cos\left(\overrightarrow{BA}\hat{}\overrightarrow{CD}\right)$

$= \frac{\overrightarrow{BA}.\overrightarrow{CD}}{|\overrightarrow{BA}||\overrightarrow{CD}|} = \frac{-\vec{b}.(\vec{d}-\vec{c})}{|\vec{b}||\vec{d}-\vec{c}|}$ ...(1)



Now numerator of (1) = $\vec{b}.\vec{c}-\vec{b}.\vec{d}$

$= |\vec{b}||\vec{c}|\cos\frac{\pi}{3}-|\vec{b}||\vec{d}|\cos\frac{2\pi}{3}$

$= (5\lambda)(\lambda)\frac{1}{2}+(5\lambda)(3\lambda)\frac{1}{2} = \frac{5\lambda^2+15\lambda^2}{2} = 10\lambda^2$

Denominator of (1) = $|\vec{b}||\vec{d}-\vec{c}|$

Now $|\vec{d}-\vec{c}|^2 = \vec{d}^2+\vec{c}^2-2\vec{c}.\vec{d}$

$9\lambda^2+\lambda^2-2(\lambda)(3\lambda)1/2 = 10\lambda^2-3\lambda^2 = 7\lambda^2$

$\therefore \quad |\vec{d}-\vec{c}| = \sqrt{7}\lambda$

Denominator of (1) = $(5\lambda)(\sqrt{7}\lambda) = 5\sqrt{7}\lambda^2$

$\therefore \quad \cos\overrightarrow{BA}\hat{}\overrightarrow{CD} = \frac{10\lambda^2}{5\sqrt{7}\lambda^2} = \frac{2}{\sqrt{7}}$

89. 28

Sol. $|\vec{u}-i|^2 = |\vec{u}|^2+1-|\vec{u}|$

$|\vec{u}-2i|^2 = |\vec{u}|^2+4-2|\vec{u}|$

So, $\left(|\vec{u}|^2+1-|\vec{u}|\right)^2 = |\vec{u}|^2\left(|\vec{u}|^2+4-2|\vec{u}|\right)$

$= |\vec{u}| = \sqrt{2}-1$

90. 8

Sol. $(x^4-1)+(y^4-1)-(z^4-1)=0$

$(x^4-1)+\lambda(y^4-1)+(z^4-1)=0$

$(x^4-1)-(y^4-1)+\lambda(z^4-1)=0$

$$\Delta=\begin{vmatrix}1 & 1 & -1\\ 1 & \lambda & 1\\ 1 & -1 & \lambda\end{vmatrix}=\lambda^2+1-\lambda+1+1+\lambda=\lambda^2+3>0$$

$\Delta_1=\Delta_2=\Delta_3=0$

$x^4-1=y^4-1=z^4-1=0$

$x=\pm1, y=\pm1, z=\pm1$

Number of solutions is 8.